

Wavelength, $6.0 \times 10^{-7} \text{ m}$

19.

Ans: D

Initially, electric force on each sphere $F = (6 \times 10^{-6})(-2 \times 10^{-6})/4\pi\epsilon_0 d^2$. [attractive]

After the spheres touch each other, the total amount of charge is $4 \times 10^{-6} \text{ C}$. Hence, each sphere will receive $2 \times 10^{-6} \text{ C}$.

Hence, $F_1 = (2 \times 10^{-6})(2 \times 10^{-6})/4\pi\epsilon_0 d^2 = F/3$ [repulsive]

20.

Ans: A